

Student Management System Using Python Report

Project Title:

**Student Management System Using
Python**

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Program: B.Tech Biotechnology

Project Number: Project 01

Domain: Python Programming

Language Used: Python 3

Abstract

The Student Management System is a menu-driven Python application developed to understand the fundamentals of programming and problem-solving. The project demonstrates how Python can be used to perform basic CRUD (Create, Read, Update, Delete) operations using dictionaries, functions, loops, and conditional statements. The application allows users to add student records, display student details, search for individual students, update existing records, and delete records efficiently.

This project strengthened fundamental programming concepts including data structures, function design, menu-driven programming, user input handling, and logical thinking. The project serves as a foundation for developing larger software applications and prepared the programmer for future bioinformatics software development using Python.

Keywords: Python, Student Management System, CRUD Operations, Functions, Dictionary, Programming Fundamentals.

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1. Introduction

Python is one of the most widely used programming languages because of its simplicity, readability, and extensive library support. Learning Python is an essential first step before entering the field of bioinformatics. This project demonstrates how Python can be used to manage student information through a simple menu-driven application.

2. Objective

The objectives of this project are:

To understand Python programming fundamentals.

To implement CRUD operations.

To practice functions and loops.

To manage data using dictionaries.

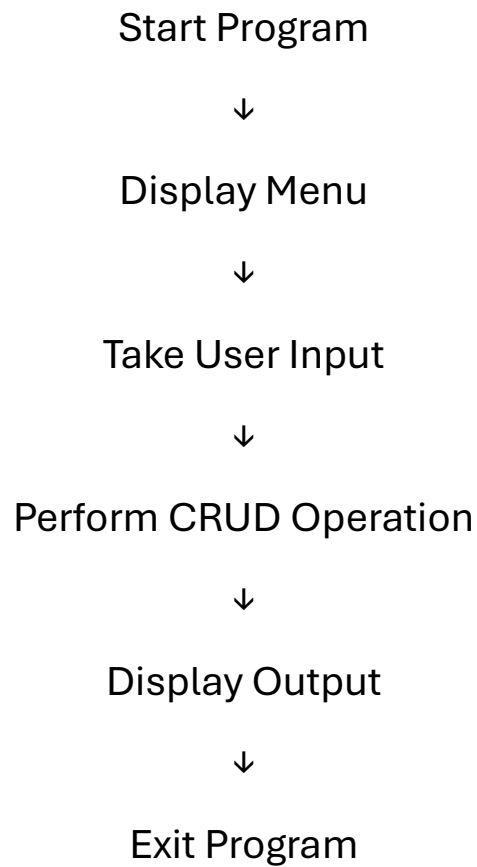
To improve logical problem-solving skills.

3. Software Requirements

Software	Purpose
Python3	Programming
VS Code	Code Editor
Terminal	Program Execution

4. Methodology

Workflow



5. Project Features

- Add Student
- View Students
- Search Student
- Update Student
- Delete Student
- Exit Program

6. Results

The Student Management System successfully performed all CRUD operations without errors. Student information could be stored, retrieved, modified, and deleted through a user-friendly menu-driven interface.

7. Discussion

The project improved understanding of dictionaries, loops, functions, conditional statements, and user input handling. These concepts form the programming foundation required for future bioinformatics software development.

8. Conclusion

The Student Management System successfully demonstrated the implementation of Python fundamentals in a real-world application. The project enhanced logical thinking and programming confidence and established the foundation required for advanced Python and bioinformatics projects.

9. References

Python Official Documentation

<https://docs.python.org/3/>

Visual Studio Code

<https://code.visualstudio.com/>